

Multiple Choice (10 marks)

1. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$
2. If your training loss increases with number of epochs, which of the following could be a possible issue with the learning process?
 - (a) Regularization is too low and model is overfitting
 - (b) Regularization is too high and model is underfitting
 - (c) Step size is too large
 - (d) Step size is too small
3. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
4. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships
5. Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs.
Statement 2| The ResNet-50 model has over 1 billion parameters.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Clustering is an effective method for detecting fraudulent credit card transactions by grouping similar spending patterns.
7. True or False: The joint probability distribution $P(X, Y, Z)$ can be expressed as $P(X) * P(Y) * P(Z)$.
8. True or False: Out-of-distribution detection is also referred to as anomaly detection in machine learning.
9. True or False: The number of test examples needed for statistically significant results is larger if the error rate is smaller.
10. True or False: The dimensionality of the null space of the matrix A is 2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find average value of the numbers in a given tuple of tuples.
12. Write a function to check whether the given key is present in the dictionary or not.

End of Paper